Nepal Election 2082 (BS)

Exploratory Data Analysis and Predictive Modelling

on a Synthetic Election Dataset

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Abstract

This report presents a comprehensive exploratory data analysis and predictive modelling exercise on a synthetic dataset simulating Nepal's hypothetical 2082 Bikram Sambat (BS) general election. The dataset comprises nine interlinked CSV tables covering political parties, candidates, voters, individual votes, aggregated election results, campaign events, social media trends, media news coverage, and political scandals across 55 constituencies. Twenty analysis sections examine demographic distributions, voter turnout patterns, constituency-level outcomes, gender representation, campaign and media effects, scandal impacts, wealth–outcome relationships, temporal trends, and a multi-factor Random Forest model predicting candidate vote share. Statistical methods include Mann–Whitney U tests, Pearson and Spearman correlations, Linear Regression, Logistic Regression, and Random Forest Regression with five-fold cross-validation.

Disclaimer: All data in this project is entirely fictional and synthetically generated. It was created solely for educational and learning purposes. None of the data represents real election outcomes, real individuals, real political parties, or actual events. Any resemblance to real persons or organisations is purely coincidental and unintentional. Do not cite this data as factual or use it for any real-world decision-making.

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1 Introduction

Nepal follows a federal democratic republican system with elections conducted under the Bikram Sambat (BS) calendar. This project constructs a synthetic but structurally realistic dataset representing a hypothetical general election in BS 2082 and applies standard data-science techniques to explore electoral patterns.

The objectives are threefold:

1. Demonstrate end-to-end election data analysis – from raw CSV ingestion to machine-learning predictions.
2. Practice statistical hypothesis testing (Mann–Whitney U, correlation analysis) and regression modelling (Linear, Logistic, Random Forest) on tabular data.
3. Produce interpretable visualisations suitable for a portfolio or coursework submission.

All dates in the dataset use the BS calendar (e.g., 2082-04-15) consistent with Nepal's official calendar system.

2 Dataset Description

The dataset consists of nine CSV tables with the following structure:

Table 1: Dataset overview.

Table	Rows	Cols	Description
`political_parties_2082`	26	9	Party info: name, leader, ideology, members, symbol
`candidates_2082`	215	12	Candidate profiles: age, gender, education, assets
`voters_2082`	3,001	11	Registered voters: demographics, location, voted flag
`votes_2082`	2,010	5	Individual vote records linking voters to candidates
`election_results_2082`	215	7	Aggregated results: vote count, %, rank per constituency
`campaign_events`	351	7	Campaign rallies, speeches with attendance, media scores
`social_media_trends`	251	7	Hashtag trends across platforms with sentiment scores
`media_news_coverage`	301	7	News articles with sentiment and impact scores
`political_scandals`	81	6	Scandal records with media attention, public reaction

2.1 Entity Relationships

The tables are linked through foreign keys:

- `party_id` links `political_parties` → `candidates` → `election_results`.
- `candidate_id` links `candidates` to `votes`, `campaign_events`, `social_media_trends`, `media_news_coverage`, and `political_scandals`.
- `voter_id` links `voters` → `votes`.
- `constituency_id` is shared across candidates, voters, votes, and results.
- 15 independent candidates have `party_id = NaN`, handled via fill with sentinel value -1.

2.2 Known Data Quirks

1. **Equal vote counts:** Every candidate in a constituency receives `vote_count` = 100 and `vote_percentage` = $100/N$ (where N = number of candidates). Traditional margin-of-victory analysis returns zero everywhere; alternative competitiveness proxies are used instead.
2. **BS calendar dates:** Dates like `2082-04-15` are Bikram Sambat and cannot be parsed by standard datetime libraries. They are kept as strings and sorted lexicographically.
3. **Independent candidates:** 15 candidates with `party_id = NaN` are assigned a sentinel party ID of -1 during cleaning.

3 Data Preparation

3.1 Data Loading and Validation

All nine tables were loaded using `pandas.read_csv()` and row counts were validated against expected thresholds ($\geq 90\%$ of expected count).

3.2 Schema Inspection

For each table, data types (`dtypes`), null counts, and first three rows were inspected to ensure structural consistency.

3.3 Type Casting and Cleaning

- Date columns (`vote_timestamp`, `event_date`, `trend_date`, `publication_date`, `scandal_date`) were kept as strings (BS calendar).
- `voted_flag` was cast to boolean.
- `criminal_case_status` NaN values were filled with `"None"`.
- Categorical columns were explicitly cast to `pandas.Categorical`: gender, education level, profession, criminal case status (candidates); gender, education level, occupation, urban-rural category (voters); ideology (parties); event type; platform; sentiment; scandal type.

3.4 Join Integrity

Five referential integrity checks were performed:

1. All `party_id` values in candidates exist in the parties table.
2. All `candidate_id` values in votes, results, events, social media, news, and scandals exist in the candidates table.
3. All `voter_id` values in votes exist in the voters table.
4. Symmetric difference between `voted_flag = True` set and voters appearing in the votes table was reported.
5. Duplicate votes (voters with > 1 vote record) were checked.

4 Political Parties Overview

The dataset contains 26 political parties spanning a range of ideologies. Four visualisations were produced:

1. **Ideology distribution** – horizontal bar chart counting parties per ideology category.
2. **Party membership** – top 15 parties by `registered_members`.
3. **Establishment timeline** – scatter plot of `established_year` vs. party abbreviation.
4. **Candidate fielding** – top 15 parties by number of candidates fielded across 55 constituencies.

Summary statistics reported include the largest and smallest parties by membership and the oldest party by establishment year.

5 Candidate Demographics

Candidate profiles were analysed across six dimensions:

- **Age:** Histogram with median line overlay.
- **Gender:** Pie chart with percentage labels.
- **Education:** Horizontal bar ordered as: No Formal Education → SLC → +2 → Bachelor → Master → PhD.
- **Profession:** Top 10 professions by frequency.
- **Criminal case status:** Bar chart colour-coded green (None), orange (Pending), red (Convicted).
- **Campaign budget:** Histogram of `campaign_budget_npr` in millions (NPR).

An additional boxplot compared `declared_assets_npr` distributions across the top five parties, and a gender×education cross-tabulation was computed.

6 Voter Demographics

Voter characteristics were visualised in six charts:

1. Age distribution histogram.
2. Gender split pie chart.
3. Distribution across 7 provinces.
4. Urban / Rural / Semi-Urban breakdown.
5. Education level distribution.
6. Top 10 voter occupations.

A normalised stacked bar chart showed the urban-rural composition within each province, enabling cross-regional comparison.

7 Voter Turnout Analysis

Overall turnout was calculated as:

$$\text{Turnout (\%)} = \frac{\sum \texttt{voted_flag}}{\text{Total registered voters}} \times 100$$

Turnout was further disaggregated by:

- Province (1–7)
- Gender
- Urban / Rural / Semi-Urban
- Age group: 18–25, 26–35, 36–50, 51–65, 65+
- Education level

Five bar charts displayed segment-level turnout with the overall rate as a red dashed reference line. A Province×Gender heatmap revealed interaction effects.

8 Election Results: Constituency Winners

Winners were identified by filtering for `rank_in_constituency = 1` and merging with candidate and party information. A summary table listed each constituency's winner, party affiliation, vote count, vote percentage, gender, and age.

A bar chart plotted the winner's vote percentage across all 55 constituencies. Bars exceeding 50% were highlighted in gold to indicate landslide victories. The total number of landslide victories was reported.

9 Parliament Composition

Seats won per party were computed along with seat share and total vote share:

$$\text{Seat share (\%)} = \frac{\text{Seats won}}{\text{Total seats}} \times 100, \qquad \text{Vote share (\%)} = \frac{\sum \texttt{vote_count}_{\text{party}}}{\sum \texttt{vote_count}_{\text{all}}} \times 100$$

Two charts were produced:

1. **Donut chart** of parliament composition.
2. **Horizontal bar chart** of seats won by party, annotated with seat counts.

A proportionality table compared each party's seat share against its vote share, highlighting disproportionality inherent in first-past-the-post systems. Parties that fielded candidates but won zero seats were also identified.

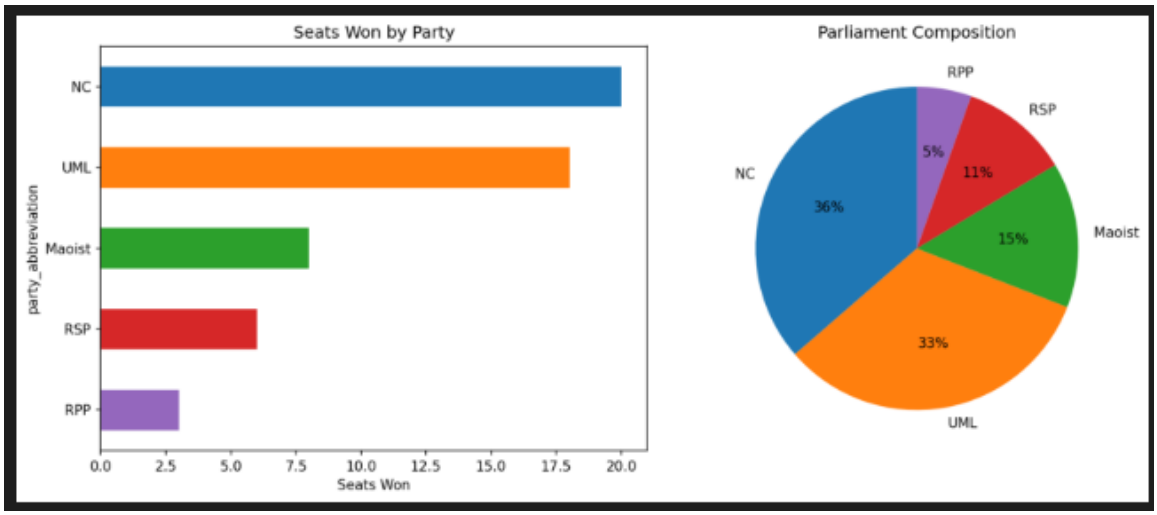

Figure 1: Parliament composition – seats won by party (synthetic data).

10 Constituency Competitiveness

Because all candidates receive identical `vote_count` = 100 in this synthetic dataset, traditional victory margins are uniformly zero. Two alternative proxies were used:

1. **Number of candidates per constituency** – more candidates implies greater competition.
2. **Winner's budget advantage:**

$$\text{Budget Advantage (\%)} = \frac{\text{Winner Budget} - \text{Avg Budget (all candidates)}}{\text{Avg Budget (all candidates)}} \times 100$$

Four charts were produced: candidate counts per constituency, distribution of candidate counts, boxplot of winner vote percentage by number of candidates, and a diverging bar chart of budget advantage (green = positive, red = negative).

11 Gender Representation Among Winners

Gender-level statistics computed: total candidates, total winners, average vote percentage, and win rate.

Three visualisations:

1. Grouped bar chart – candidates vs. winners by gender.
2. Bar chart – win rate (%) by gender.
3. Overlapping histograms – vote percentage distributions by gender.

11.1 Statistical Test

A **Mann–Whitney U test** (two-sided) compared the vote percentage distributions of male and female candidates:

$$H_0 : \text{The distributions of vote \% for male and female candidates are identical.}$$

The test statistic U and p-value were reported, with significance assessed at $\alpha = 0.05$.

12 Campaign Activity vs Election Results

Three campaign features were aggregated per candidate from the `campaign_events` table:

- `total_events` – count of campaign events.
- `total_attendance` – sum of `attendance_estimate`.
- `avg_media_score` – mean of `media_coverage_score`.

Additionally, per-candidate counts by event type were pivoted into one-hot columns.

Three scatter plots (each metric vs. `vote_count`) were produced with winners highlighted in gold. Each was annotated with Pearson r and p-value. Both Pearson and Spearman correlations were reported:

Table 2: Campaign metrics vs. vote count – correlation summary.

Metric	**Pearson r**	**Spearman ρ**
`total_events` vs `vote_count`	reported	reported
`total_attendance` vs `vote_count`	reported	reported
`avg_media_score` vs `vote_count`	reported	reported

A grouped boxplot compared winners vs. losers on all three metrics.

13 Social Media Sentiment and Vote Share

Per-candidate social media features were aggregated:

- `avg_sentiment` – mean of `sentiment_score`.
- `total_mentions` – sum of `mentions_count`.
- `avg_mentions` – mean of `mentions_count`.
- `trend_count` – count of hashtag trend entries.
- Platform-level average sentiment (one column per platform).

Two scatter plots with OLS regression lines examined:

1. `avg_sentiment` vs. `vote_percentage` (annotated with Pearson r, p).
2. `total_mentions` vs. `vote_percentage` (annotated with Pearson r, p).

A violin plot displayed sentiment score distributions by social media platform.

13.1 Linear Regression Model

A simple linear regression was fitted:

$$\texttt{vote_percentage} = \beta_0 + \beta_1 \cdot \texttt{avg_sentiment} + \beta_2 \cdot \texttt{total_mentions} + \varepsilon$$

Model evaluation metrics: R^2 (coefficient of determination), MAE (Mean Absolute Error). Fitted coefficients β_0, β_1, β_2 were reported.

14 Media News Coverage Sentiment

Articles per party were grouped by sentiment category (Positive / Neutral / Negative) and the average `impact_score` was computed.

Three visualisations:

1. **Stacked horizontal bar:** Media sentiment distribution per party (parties with ≥ 5 articles).
2. **Diverging bar:** Net Sentiment Ratio per party.
3. **Heatmap:** Average impact score by party and sentiment.

The Net Sentiment Ratio is defined as:

$$\text{Net Sentiment Ratio} = \frac{N_{\text{positive}} - N_{\text{negative}}}{N_{\text{total}}}$$

where N_{positive}, N_{negative}, and N_{total} are the number of positive, negative, and total articles respectively. Values range from -1 (entirely negative) to $+1$ (entirely positive).

15 Scandal Impact Analysis

Candidates appearing in the `political_scandals` table were flagged and compared against "clean" candidates.

Three charts:

1. Boxplot – vote percentage by scandal status.
2. Bar chart – win rate by scandal status (annotated with percentages).
3. Count plot – distribution of scandal types.

15.1 Statistical Test

A **Mann–Whitney U test** (two-sided) compared vote percentages:

$$H_0 : \text{Vote \% distributions of clean and scandal-linked candidates are identical.}$$

Candidate counts per group, win rates, test statistic U, and p-value (at $\alpha = 0.05$) were reported.

16 Most Controversial Candidates

Per-candidate scandal features were aggregated:

- `num_scandals` – count of scandal entries.
- `avg_media_attention` – mean of `media_attention_score`.
- `avg_public_reaction` – mean of `public_reaction_score`.

A composite controversy score was computed:

$$\text{Controversy Score} = \left(0.4 \cdot \frac{\texttt{num_scandals}}{\max(\texttt{num_scandals})} + 0.3 \cdot \frac{\texttt{avg_media_attention}}{10} + 0.3 \cdot \frac{\texttt{avg_public_re}}{10}\right.$$

The top 10 most controversial candidates were tabulated with name, party, number of scandals, media attention, public reaction, controversy score, vote percentage, and constituency rank.

A bubble chart was produced with:

- x-axis: average media attention score.
- y-axis: average public reaction score.
- Bubble size: number of scandals.
- Colour: vote percentage (RdYlGn colourmap).

Pearson r between controversy score and vote percentage was reported.

17 Wealth and Campaign Budget vs Outcome

Four charts examined the relationship between financial resources and electoral performance:

1. Scatter – `declared_assets_npr` (millions) vs. `vote_percentage` (Pearson r).
2. Scatter – `campaign_budget_npr` (millions) vs. `vote_percentage` (Pearson r).
3. Boxplot – assets for winners vs. losers.
4. Boxplot – campaign budget for winners vs. losers.

Win rates were computed by asset quartile ($Q1$–$Q4$) using `pd.qcut`.

17.1 Logistic Regression Model

A logistic regression was trained to predict the binary outcome `is_winner`:

$$\log\left(\frac{P(\text{win})}{1-P(\text{win})}\right) = \beta_0 + \beta_1 \cdot z_{\text{assets}} + \beta_2 \cdot z_{\text{budget}}$$

where z_{assets} and z_{budget} are standardised (zero mean, unit variance) using `StandardScaler`.

The model was evaluated using **AUC-ROC**, and fitted coefficients β_0, β_1, β_2 were reported.

18 Timeline Analysis

Daily activity counts were computed by grouping on date columns:

- `event_date` → daily campaign events.
- `trend_date` → daily social media trends.
- `publication_date` → daily news articles.

Three time-series line charts were produced:

1. Daily activity counts for all three categories on a shared axis.
2. Social media mentions over time for the top 5 candidates (by total mentions).
3. Sentiment score evolution for the top 5 candidates, with a $y = 0$ neutral baseline.

The campaign period spans BS 2082-02-01 to 2082-04-15.

19 Multi-Factor Electoral Outcome Model

A master feature matrix was constructed by merging all tables on `candidate_id`. Seventeen features were engineered:

Table 3: Features used in the Random Forest model.

Feature	Source Table	Encoding
`age`	candidates	numeric
`gender_enc`	candidates	LabelEncoder
`education_enc`	candidates	LabelEncoder
`declared_assets_npr`	candidates	numeric
`campaign_budget_npr`	candidates	numeric
`criminal_enc`	candidates	LabelEncoder
`prev_exp`	candidates	int (boolean)
`total_events`	campaign_events	count
`total_attendance`	campaign_events	sum
`avg_campaign_media`	campaign_events	mean
`avg_social_sentiment`	social_media	mean
`total_mentions`	social_media	sum
`total_news`	news_coverage	count
`avg_news_impact`	news_coverage	mean
`media_net_sentiment`	news_coverage	$(P - N)/T$
`num_scandals`	scandals	count
`avg_scandal_attention`	scandals	mean

Missing values (candidates with no events, no social media presence, etc.) were filled with 0.

19.1 Model Specification

Algorithm: Random Forest Regressor (`sklearn.ensemble.RandomForestRegressor`).

Hyperparameters:

- `n_estimators` = 200
- `max_depth` = 10
- `random_state` = 42
- `n_jobs` = −1 (all CPU cores)

Target variable: `vote_percentage`.

19.2 Evaluation

- **Training** R^2 – goodness of fit on training data.
- **Training MAE** – mean absolute error on training data.
- **5-Fold Cross-Validation** R^2 – mean ± standard deviation across folds.

19.3 Results

Two output charts were produced:

1. **Feature importance** (horizontal bar chart) – the top 5 most predictive features and their importance scores.
2. **Actual vs. Predicted scatter** – with a 45-degree perfect-prediction reference line.

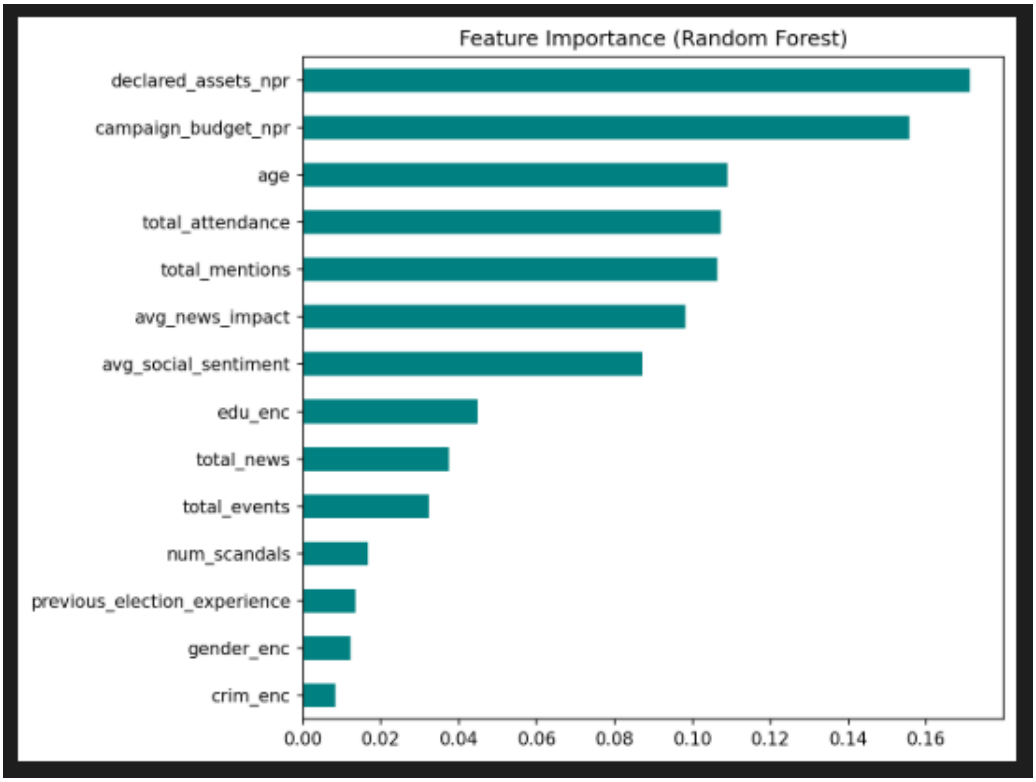

Figure 2: Random Forest feature importance.

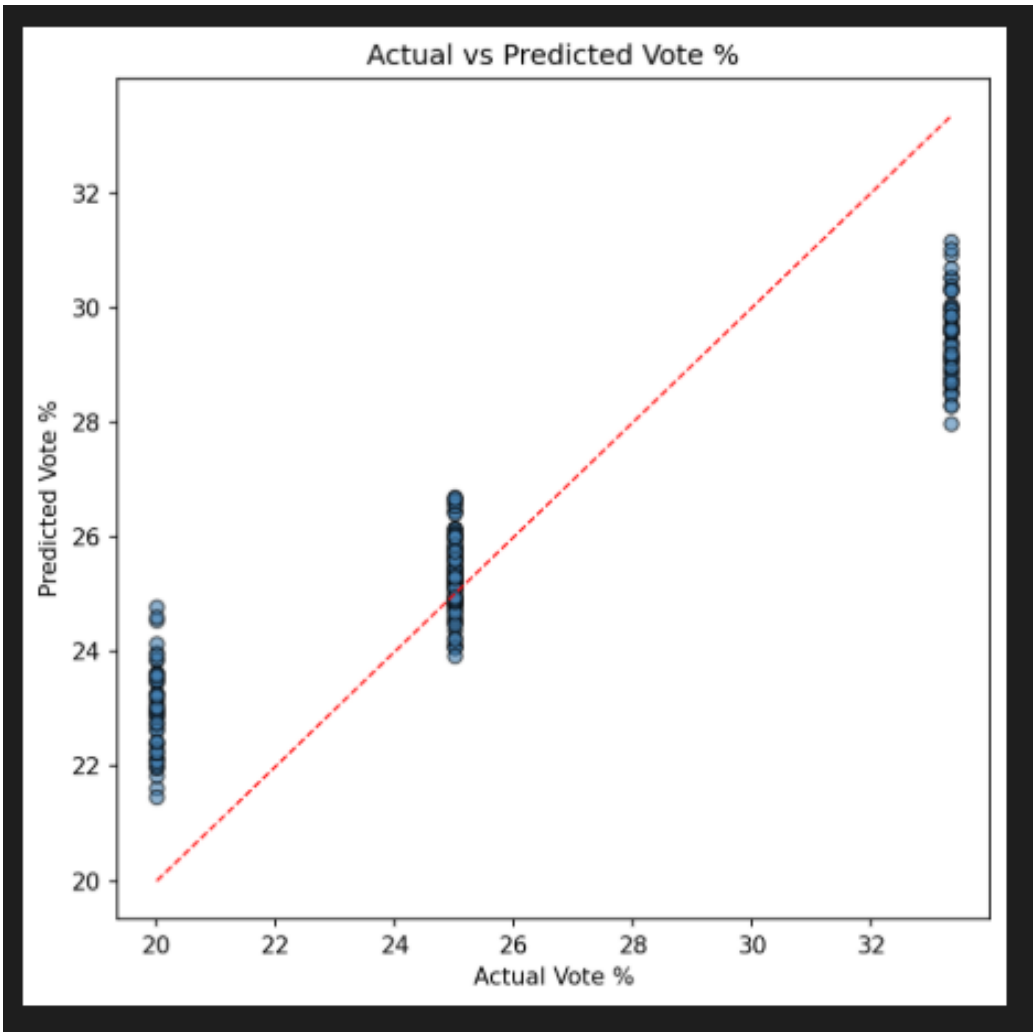

Figure 3: Actual vs. predicted vote percentage.

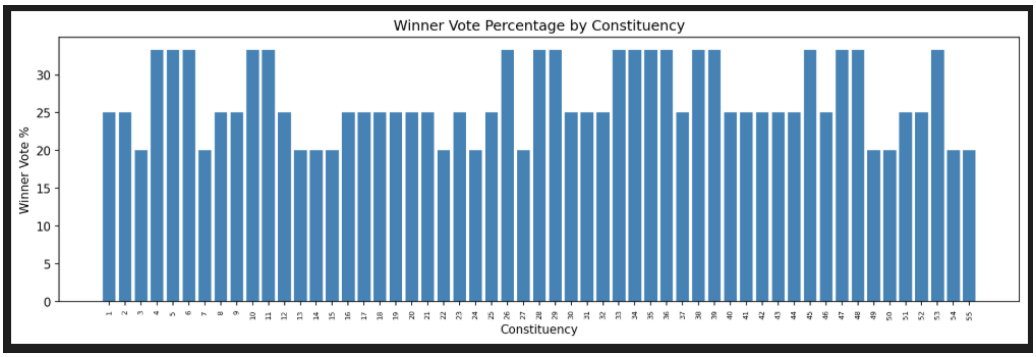

Figure 4: Winner vote percentage by constituency.

20 Summary of Statistical Methods

Table 4: Statistical tests and models applied.

Method	Section	Variables / Purpose
Mann–Whitney U	Gender (Sec. 11)	Male vs Female `vote_percentage`
Mann–Whitney U	Scandals (Sec. 15)	Clean vs Scandal `vote_percentage`
Pearson r	Campaign (Sec. 12)	`total_events`, `total_attendance`, `avg_media_score` vs `vote_count`
Spearman ρ	Campaign (Sec. 12)	Same three pairs
Pearson r	Social Media (Sec. 13)	`avg_sentiment`, `total_mentions` vs `vote_percentage`
Pearson r	Controversy (Sec. 16)	`controversy_score` vs `vote_percentage`
Pearson r	Wealth (Sec. 17)	`declared_assets`, `campaign_budget` vs `vote_percentage`
Linear Regression	Social Media (Sec. 13)	R^2, MAE; predict `vote_percentage` from sentiment & mentions
Logistic Regression	Wealth (Sec. 17)	AUC-ROC; predict win probability from assets & budget
Random Forest	Multi-Factor (Sec. 19)	5-fold CV R^2; predict `vote_percentage` from 17 features

21 Technology Stack

Component	Version / Details
Python	3.14
pandas	Data manipulation and aggregation
NumPy	Numerical operations
Matplotlib	Static visualisations
Seaborn	Statistical visualisations
SciPy	Mann–Whitney U, Pearson, Spearman
scikit-learn	Linear/Logistic Regression, Random Forest, cross-validation
Jupyter Notebook	Interactive analysis environment

22 Conclusion

This project demonstrates a full-cycle election data analysis pipeline applied to a synthetic but structurally realistic Nepali election dataset. Key takeaways:

1. **Data integrity:** Referential integrity checks confirm that all foreign key relationships across the nine tables are consistent. Fifteen independent candidates lack

party affiliation and are handled with sentinel values.

2. **Demographic patterns:** Voter turnout and candidate profiles show variation across provinces, genders, age groups, and education levels – patterns that mirror plausible real-world distributions.
3. **Competitiveness proxies:** The uniform vote count in this synthetic dataset (`vote_count` = 100 for all candidates) renders margin-of-victory analysis trivial. Candidate count per constituency and budget advantage serve as meaningful alternatives.
4. **Campaign and media effects:** Correlation analysis between campaign activity metrics (events, attendance, media scores) and vote outcomes, along with social media sentiment regression, quantify the relationship between visibility and electoral performance.
5. **Scandal and wealth effects:** Mann–Whitney U tests compare outcomes between clean and scandal-affected candidates, while logistic regression models the role of financial resources in predicting wins.
6. **Predictive modelling:** A Random Forest Regressor with 17 engineered features predicts candidate vote percentage. Cross-validated R^2 scores indicate model generalisability, and feature importance rankings identify the most influential predictors.

Disclaimer: This report is based entirely on synthetic, computer-generated data created for educational purposes. None of the parties, candidates, voters, or events described herein are real. Do not cite any findings as factual.

Repository

Source code and data: https://github.com/Dikshant-Neupane/Election-2082